

Credit Card Fraud Detection Analysis – Detailed Project Report

This report documents the complete workflow followed to build the Credit Card Fraud Detection Analysis project. The project includes a Python-based ETL pipeline and a Tableau dashboard to analyze fraud patterns in credit card transaction data. Dataset size: 284,807 transactions Fraud transactions: 492 Fraud percentage: ~0.17%

1. Technologies Used

- 1 Python – Used for building the ETL pipeline
- 2 Pandas – Used for loading and analyzing the dataset
- 3 Tableau – Used for building the interactive dashboard
- 4 Git & GitHub – Used for version control and project hosting

2. Project Folder Structure

```
credit-card-fraud-detection
├── data
│   └── creditcard.csv
├── scripts
│   ├── extract.py
│   ├── transform.py
│   ├── load.py
│   └── etl_pipeline.py
├── dashboard
│   └── fraud_detection_dashboard.twbx
└── README.md
```

3. Extract Step (extract.py)

```
import pandas as pd

def extract_data():
    df = pd.read_csv("data/creditcard.csv")
    print("Dataset loaded successfully")
    print(df.head())
    return df
```

Explanation: 1. import pandas as pd - Imports the Pandas library used for data analysis. 2. pd.read_csv() - Loads the credit card dataset from the data folder. 3. df.head() - Displays the first five rows to verify the dataset loaded correctly. 4. return df - Returns the dataframe to be used in the transformation stage.

4. Transform Step (transform.py)

```
def transform_data(df):
```

```

print("Dataset shape:", df.shape)

print("Fraud vs Normal Transactions:")
print(df['Class'].value_counts())

return df

```

Explanation: 1. df.shape - Displays the number of rows and columns in the dataset. 2. df['Class'].value_counts() - Counts the number of normal (0) and fraud (1) transactions. 3. This step helps understand the imbalance in the dataset.

5. Load Step (load.py)

```

def load_data(df):
    df.to_csv("data/processed_data.csv", index=False)
    print("Processed dataset saved successfully")

```

Explanation: 1. df.to_csv() - Saves the processed dataset. 2. index=False - Prevents Pandas from writing row index numbers to the file. 3. The processed dataset will later be used in Tableau for visualization.

6. ETL Pipeline Script (etl_pipeline.py)

```

from extract import extract_data
from transform import transform_data
from load import load_data

def run_pipeline():
    print("Starting ETL Pipeline...")

    df = extract_data()
    df = transform_data(df)
    load_data(df)

    print("ETL Pipeline completed successfully")

if __name__ == "__main__":
    run_pipeline()

```

Explanation: 1. The ETL pipeline coordinates all three steps. 2. Extract stage loads the raw dataset. 3. Transform stage analyzes the dataset and inspects fraud distribution. 4. Load stage exports the processed dataset. 5. The pipeline runs when the script is executed directly.

7. Tableau Dashboard Creation

- 1 Import the processed dataset into Tableau.
- 2 Create a bar chart showing Fraud vs Normal transactions.
- 3 Create a scatter plot showing Transaction Amount Distribution.
- 4 Create a line chart showing Fraud Transactions Over Time.
- 5 Combine all visualizations into a single dashboard.
- 6 Publish the dashboard to Tableau Public.

8. Key Insights

- 1 Fraud transactions represent only ~0.17% of total transactions.
- 2 Fraud events appear in clusters over time.
- 3 Fraud transactions show higher variability in transaction amounts.